

CLOUD CART

Convenience for Every Purchase,
Confidence in Every Order.



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AGENDA



Problem Statement



Solution Overview



Key Features



AWS Services Used



Architecture



Future Work



Conclusion

Problem Statement



- ▶ Traditional e-commerce platforms face challenges such as **infrastructure management**, **scalability limitations**, **high operational costs**, and **security concerns**. These issues impact application performance and user experience. CloudCart solves these challenges by providing a **secure**, **scalable**, and fully **serverless** e-commerce platform using **AWS services**.

SOLUTION OVERVIEW



- ▶ Developed **CloudCart**, a cloud-native e-commerce platform using **AWS Serverless Architecture**.



- ▶ Enables users to **register, log in, browse products, add items** to the cart, and **place orders** through a simple and responsive interface.



- ▶ Uses **Amazon S3** for website hosting and **CloudFront** for fast, secure content delivery.



- ▶ Integrates **Amazon Cognito** for secure user authentication and authorization.



- ▶ Employs **API Gateway, AWS Lambda, and Amazon DynamoDB** to provide scalable backend services and efficient data management.



- ▶ Delivers a **secure, scalable, highly available**, and **cost-effective** online shopping experience with minimal infrastructure management.

KEY FEATURES



- ▶ Secure user registration and login using **Amazon Cognito**.



- ▶ Browse products with a **simple and responsive** user interface.



- ▶ **Add, update, and remove** items from the shopping cart.



- ▶ Place orders securely with **real-time order processing**.



- ▶ Serverless backend powered by **AWS Lambda** and **API Gateway**.



- ▶ Fast and scalable data storage using **Amazon DynamoDB**.



- ▶ Static website hosting with **Amazon S3** and global content delivery through **CloudFront**.



- ▶ Automatic scalability with **no server management** required.



- ▶ High **availability**, enhanced **security**, and **cost-effective** cloud infrastructure.

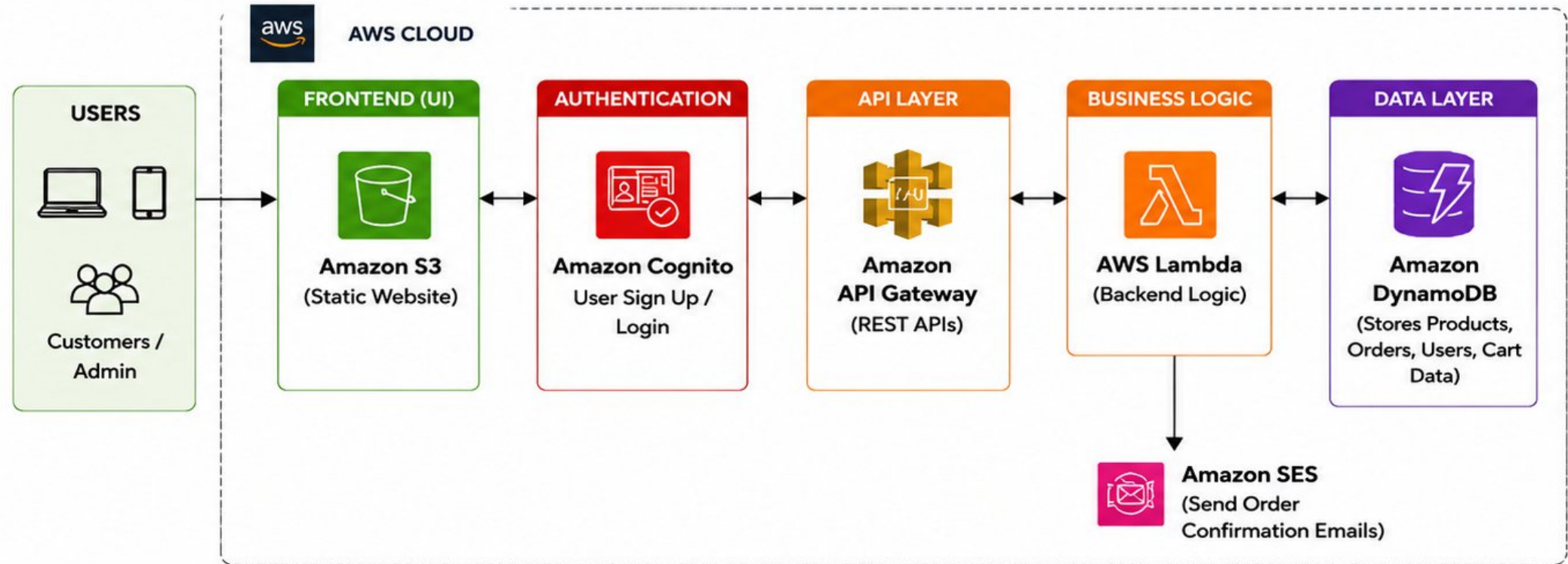
AWS Services used

AWS SERVICES USED		
AWS Services		Purpose
	Amazon S3	Stores and hosts the static website (frontend)
	Amazon Cognito	Handles user registration, login, authentication and authorization
	Amazon DynamoDB	Stores products, cart items, orders and user data
	AWS Lambda	Executes backend logic and processes business requests
	Amazon API Gateway	Exposes RESTful APIs and routes requests to Lambda functions
	Amazon SES	Sends order confirmation and notification emails to users

ARCHITECTURE

ARCHITECTURE

CloudCart – Serverless E-Commerce Application



AWS SERVICES USED


Amazon S3
(Static Frontend)


Amazon Cognito
(Authentication)


Amazon DynamoDB
(Database)


AWS Lambda
(Backend)


Amazon API Gateway
(API Management)


Amazon SES
(Email Service)

FUTURE WORK



- ▶ Integrate secure online **payment gateways** (UPI, cards, net banking).



- ▶ Implement real-time **order tracking** and delivery status updates.



- ▶ Add **AI-based** product recommendation and personalized shopping features.



- ▶ Develop an **inventory management** dashboard for administrators.



- ▶ Introduce customer **analytics** and sales reporting dashboards.



- ▶ Enable **multi-vendor** support and product management.



- ▶ Build a **mobile application** for Android and iOS users.



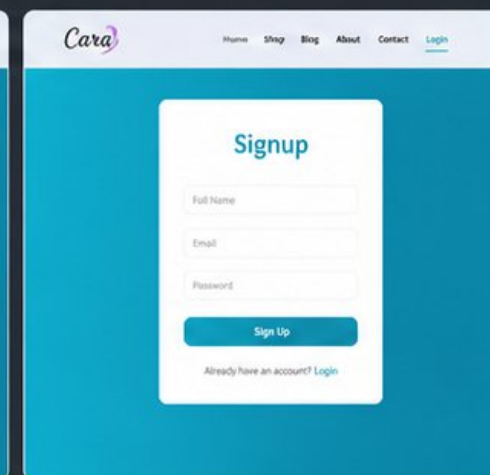
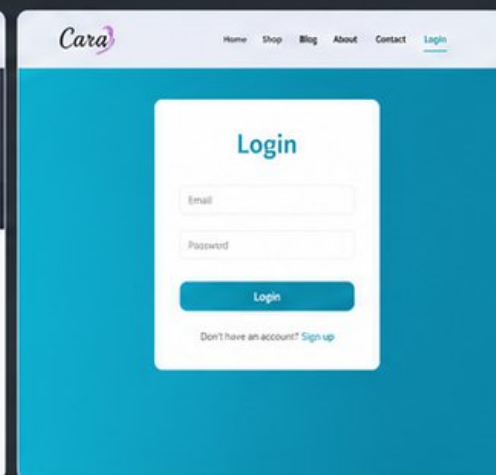
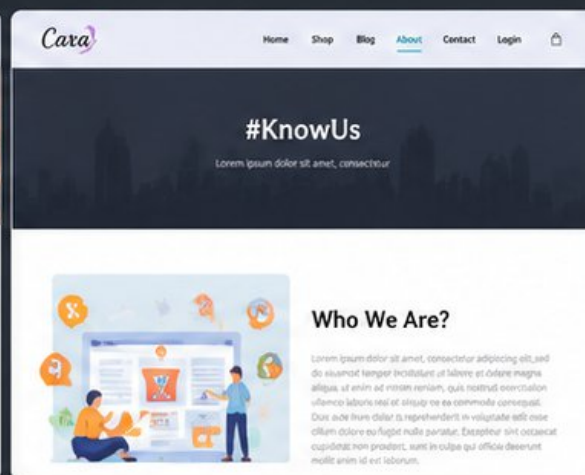
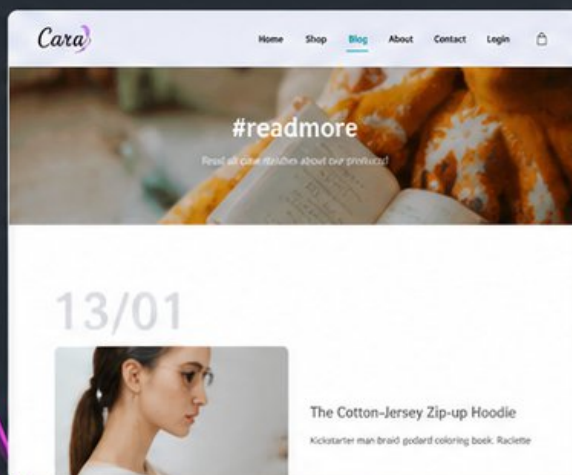
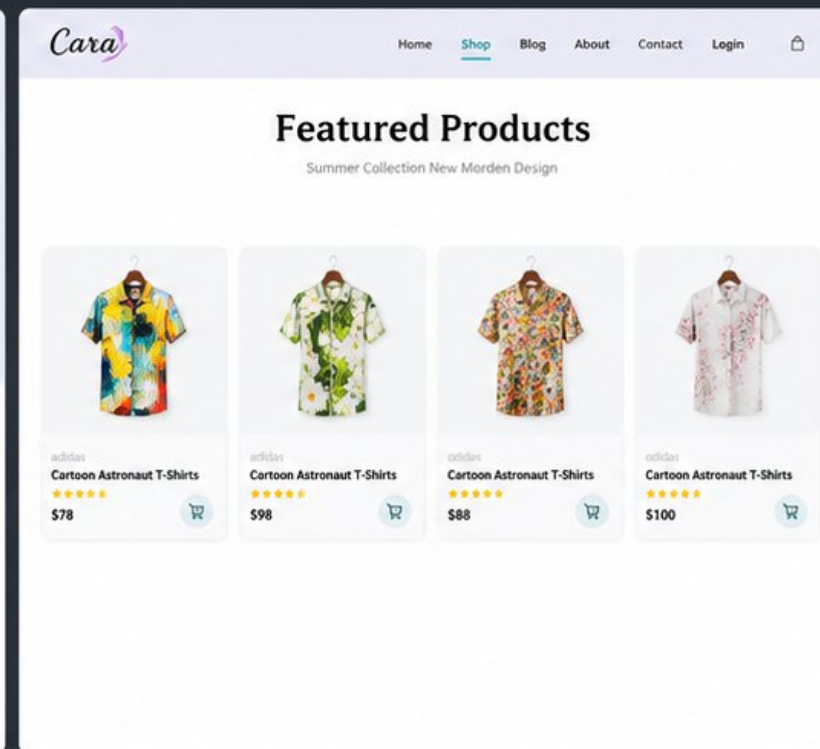
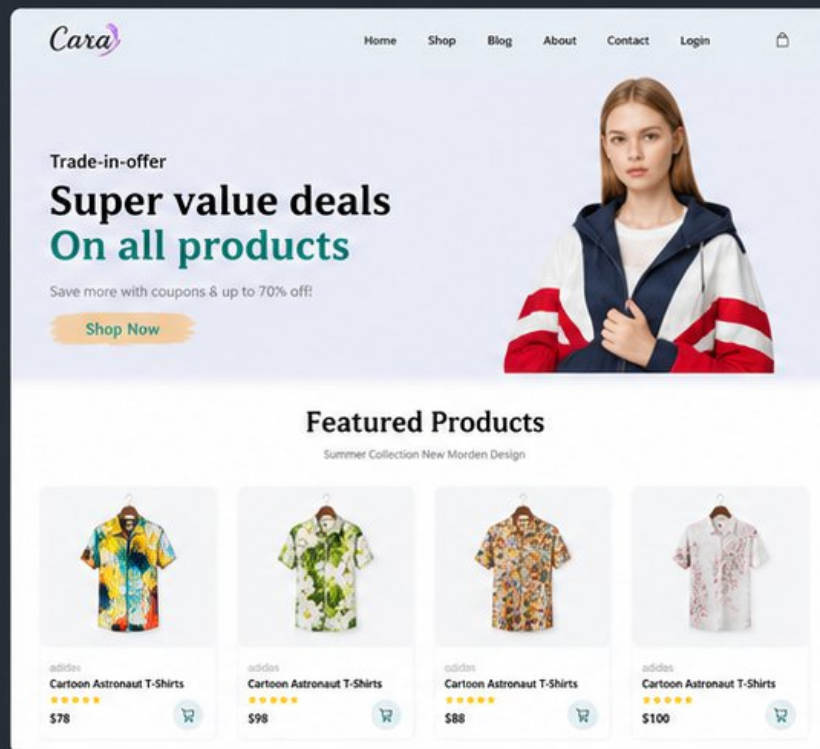
- ▶ Implement advanced **search, filtering**, and **wishlist** functionality.



- ▶ Add **notification services** for orders, offers, and cart reminders.

WEBSITE OUTPUT

- Snapshots of key pages including Home, Shop, Blog, About, Login and Signup.



CONCLUSION



- ▶ Developed a **secure** and **scalable** serverless e-commerce platform.



- ▶ Simplified infrastructure management using **AWS** cloud services.



- ▶ Delivered a **reliable, high-performance** shopping experience.



- ▶ Ensured **cost efficiency, security,** and automatic **scalability**.



- ▶ Built a foundation for **future enhancements** and **business growth**.



THANK
YOU

THANK YOU FOR
YOUR TIME AND ATTENTION